

WAITA-EACRI Joint Grant Programme

Cassava-Yam Improvement Programme (Year 2 Extension)

Proposal Version v8.1 FINAL

Submitted to: West African Agricultural Development Agency (WAADA)

Programme Officer: Adaeze Nwosu, WAADA Secretariat

Award Reference: WAADA/2026/WAITA-EACRI/Y2/007

Co-Principal Investigators

Dr. Layla McBride — National Nigerian University (NNU), Enugu — 60%
Dr. Amina Bello — East African Crop Research Institute (EACRI), Nairobi — 40%

Total Requested Award: ₦134,800,000

Period of Performance: 2026-11-01 through 2028-10-31

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1. Executive Summary

The WAITA-EACRI Cassava-Yam Improvement Programme (Year 2 Extension) seeks to scale the Year-1 baseline work conducted under Award WAADA/2026/WAITA-EACRI/Y1/004 into a two-year cross-site cultivar evaluation that pairs twelve replicated field trials at Udi LGA (Enugu State, Nigeria) with parallel laboratory characterisation at EACRI Nairobi. The programme targets a measurable lift in farm-gate beta-carotene concentration of biofortified cassava cultivars and an increase of ₦134,800,000 in farmer-cooperative throughput across the WAITA member network.

Year 1 delivered the foundation: an n=340-household baseline farmer survey in Nsukka LGA (Deliverable D-1-1), a four-cultivar variety screening with three replicates per site (D-1-2), and substantial progress on plot establishment at Udi (D-1-3) and EACRI laboratory protocol harmonisation (D-1-4). Year-2 builds on this with a cross-site evaluation (Milestone M-Y2-04, scheduled for Q3 2027) and a participatory farmer dissemination programme co-designed with the WAITA cooperative network.

This proposal requests ₦134,800,000 over twenty-four months. Funds will be administered by the NNU Office of Sponsored Programmes under the existing WAADA financial-management framework. The Co-PIs commit a combined research effort of 60% (Dr. McBride) and 40% (Dr. Bello). Detailed budget justification is provided in §7; letters of support from the WAITA Cooperative Federation and Udi LGA Agricultural Extension Office are appended as Appendix B.

2. Project Background & Scientific Rationale

2.1 The biofortification opportunity

Vitamin A deficiency remains one of the most prevalent micronutrient gaps in West and Central African subsistence diets. The Harvest+ programme has demonstrated that carotenoid-enriched (yellow-fleshed) cassava cultivars — notably TME-419, TMS-30572, and the UMUCASS-44 lineage — can shift baseline retinol-equivalent intake by an agronomically meaningful margin when grown under representative smallholder conditions and combined with appropriate post-harvest handling. The WAITA-EACRI Programme operationalises that opportunity at the level of the WAITA cooperative network: rather than testing genotypes in isolation, we couple agronomic field performance with laboratory-confirmed beta-carotene retention and with a household consumption survey designed to anchor downstream cost-benefit modelling.

2.2 Why Udi LGA, why Nsukka, why Nairobi

Udi Local Government Area is hydrologically and pedologically representative of the south-eastern Nigerian cassava belt: ferralsol soils, bimodal rainfall pattern, and a tradition of cassava-yam intercropping. The twelve experimental plots laid out in 2025 (numbered UDI-2026-001 through UDI-2026-012) carry one representative control plot — UDI-2026-070 — that sits on an outlier soil pocket and is used deliberately to bracket the on-station response surface. Nsukka LGA provides the farmer-household pool for socio-economic survey work (n=340 households surveyed in Y1), and EACRI Nairobi contributes the regional HPLC capacity for cross-site carotenoid quantification.

2.3 Relationship to the broader WAITA mandate

The WAITA Cooperative Federation has identified cassava biofortification as one of its three priority verticals for the 2026–2030 strategic period (the other two being aflatoxin reduction in stored maize and on-farm soil-fertility decision tools). This proposal aligns directly with that strategic frame, and the Year-2 outputs feed back into the Federation's annual cooperative-level planning cycle via the standing M&E; channel.

3. Year-1 Outcomes & Demonstrated Capacity

Year-1 (2025-11-01 through 2026-10-31) delivered four programmatic outputs. The status of each, as recorded in the consortium's Confluence workspace and verified against laboratory and field records on 2026-09-30, is summarised below.

ID	Deliverable	Owner	Status	Evidence anchor
D-1-1	Baseline farmer survey (Nsukka LGA, n=340)	L. McBride	Done	Airtable Farmer-Cooperative-Network: 340 r
D-1-2	Variety screening Y1 report (4 cultivars x 2 plots)	A. Bello	Done	EACRI lab report 2026-08-14
D-1-3	Field-trial plot establishment Udi (n=12)	L. McBride + D. Akpan	In Progress	Plot map 2026-09-25 / yields 14.2 etc.
D-1-4	EACRI laboratory protocol harmonisation	A. Bello	In Progress	Protocol draft v0.6 circulated 2026-09-22

Deliverables D-1-3 and D-1-4 are projected to close on 2026-10-15 and 2026-10-22 respectively, well within the Year-1 administrative grace window. The consortium therefore enters Year-2 with the full operational footprint envisaged at award kick-off — twelve established plots, a harmonised laboratory protocol, and a validated household-survey instrument.

4. Year-2 Research Plan & Objectives

4.1 Specific objectives

- O1. Quantify cross-site beta-carotene retention for four candidate cultivars (TME-419, TMS-30572, UMUCASS-44, UMUCASS-44 + biofort. backcross) under three post-harvest handling regimes (fresh, par-boiled, sun-dried gari).
- O2. Validate the Year-1 plot-establishment dataset against a second harvest cycle (2027 short rains) and extend the analysis to include sister plot UDI-2026-070 as the high-fertility control bracket.
- O3. Co-design a participatory cooperative-level dissemination programme with the WAITA Federation's Member Services Group, targeting 12 cooperatives across Enugu, Anambra and Ebonyi states.
- O4. Establish the cross-site cultivar evaluation milestone (M-Y2-04) at Q3 2027, integrating Udi field data with EACRI Nairobi laboratory output and the Nsukka household-survey panel.

4.2 Hypotheses & decision points

- H1: Mean beta-carotene retention across the four candidate cultivars, measured at harvest, will exceed 8.0 µg/g dry weight when grown under the standard Udi agronomic protocol. (Year-1 mean across the harmonised assay was 8.4 µg/g; this value is anchored at row H22 of the consortium's cassava-biofort-analysis workbook, dated 2026-09-22.)
- H2: Cross-site variance (Udi vs. EACRI on-station) will be smaller than within-site block variance, justifying the cross-site pooling assumption used in the Y2 cost-benefit model.

5. Methodology & Workplan

5.1 Field trial design

The Udi field trials (UDI-2026-001 through UDI-2026-012) use a randomised complete block design with four cultivars and three replicates per block. Plot UDI-2026-070 is included as a high-fertility outlier control and is analysed separately. Plots were sown on 2026-04-12 (Y1 short rains) and were re-sown on 2027-04-15 (Y2 schedule). Agronomic operations follow the IITA-published standard protocol for biofortified cassava trials, with site-specific modifications documented in Annex 5.A of the consortium operations manual.

5.2 Laboratory analytics

Beta-carotene quantification follows the AOAC 2005.02 HPLC method, harmonised between the NNU Crop Science laboratory and the EACRI Nairobi central laboratory during Year-1 (Deliverable D-1-4). Inter-laboratory replicates run quarterly; results are stored in the consortium's shared Airtable base (Cassava-Biofort/Field-Trial-Udi) and cross-referenced to the analytical workbook on the consortium filesystem.

5.3 Socio-economic component

The Nsukka household-survey instrument (n=340 households, validated in Year-1) will be re-administered in Year-2 with a reduced n=120 panel rotation to track intra-household consumption shifts. Survey administration is coordinated by Mr. Derek Akpan (NNU Field Office) with quality-control review by Dr. McBride.

5.4 Workplan Gantt summary

Activity	Q1 2027	Q2 2027	Q3 2027	Q4 2027	Q1 2028	Q2 2028	Q3 2028	Q4 2028
Field operations Udi (Y2 sowing → harvest)	■	■	■	■				
EACRI lab analytics	■	■	■	■	■	■		
Cross-site cultivar eval. (M-Y2-04)			■					
Household-panel rotation (n=120)		■		■				
Cooperative dissemination pilot				■	■	■		
Annual reporting to WAADA				■				■
Final report & dissemination								■

6. Personnel, Roles & Co-PI Effort

The consortium operates under a balanced Co-PI structure. For Year 2, effort is committed at 60% by Dr. McBride (NNU) and 40% by Dr. Bello (EACRI). This split reflects the Year-2 centre-of-gravity shift toward field-side operations at Udi while retaining substantial EACRI analytical capacity. Co-PI effort is reviewed and re-baselined at the Q2 2027 milestone review.

Role	Name	Institution	Effort	Function
Co-PI (lead)	Dr. Layla McBride	NNU Crop Science	60%	Field-trial lead, Nsukka survey QC, WAADA liaison
Co-PI	Dr. Amina Bello	EACRI Nairobi	40%	Lab analytics lead, cross-site harmonisation
Field officer	Mr. Derek Akpan	NNU Field Office, Udi	60%	Plot operations, agronomic data capture
Lab manager	Dr. Yetunde Olaniyan	NNU Crop Science	35%	HPLC sample throughput, lab QC
Research asst	Cassava-Y3 (TBA)	NNU — open req. F25	100%	Y3 field-assistant cohort (hiring under way)
M&E officer	Ms. Onyinye Ofili	WAITA Federation Secretariat	20%	Cooperative M&E, dissemination interface

7. Budget Justification

The total Year-2 award request is ₦134,800,000. A detailed line-item budget is provided in the companion spreadsheet (WAITA_budget_v8.1_final.xlsx), Workbook 'Year2-Budget', anchor cells B47 (sub-total ledger) and C47 (line-sum). The cell-anchored figure on row 47 matches the cover value to the kobo. Budget categories follow the standard WAADA chart of accounts.

Category	Sub-total (₦)	% of total
Personnel (Co-PIs, field officer, lab manager, RA)	51,920,000	38.5%
Field operations (Udi plots, agronomic ops, vehicles)	22,940,000	17.0%
Laboratory analytics (HPLC consumables, cross-site shipping)	19,560,000	14.5%
Travel (NNU↔EACRI, WAADA reporting, dissemination)	9,420,000	7.0%
Equipment (HPLC column refresh, field sensors)	10,200,000	7.6%
Cooperative dissemination programme	7,690,000	5.7%
Indirect costs (NNU OSP, 15% on direct)	10,930,000	8.1%
Contingency reserve	2,140,000	1.6%
TOTAL (cell B47 = C47, line-sum)	134,800,000	100.0%

Indirect costs are computed at the NNU institutional rate of 15% applied to direct costs and validated by the Office of Sponsored Programmes. The contingency reserve is held against fuel-price volatility and currency-translation slippage on imported HPLC consumables.

8. Monitoring, Evaluation & Reporting

The consortium reports quarterly to WAADA against the milestone register. The principal milestone for Year-2 is M-Y2-04 — Cross-site cultivar evaluation, scheduled for Q3 2027. The quarterly report bundle includes: (i) field-trial agronomic snapshot, (ii) laboratory analytics summary, (iii) household-panel rotation update, (iv) cooperative dissemination tracker, and (v) the financial burn-rate against the budget categories listed in §7.

Day-to-day operational records are maintained in the consortium's shared Notion workspace (Grant Tracker DB) and Airtable bases. The consortium's Confluence space (WAITA-EACRI) carries the deliverables snapshot, the Year-2 roadmap, and the milestone register. Read-only WAADA review access is granted via the WAITA Federation single-sign-on.

9. Risk Register & Mitigation

#	Risk	Likelihood	Impact	Mitigation
R1	Rainfall delay disrupts Udi sowing window	Medium	Medium	Pre-staged inputs + 2-week sowing flex
R2	HPLC column failure halts assay throughput	High	High	Budgeted column refresh + EACRI fallback
R3	FX slippage on imported reagents	Medium	Medium	Contingency reserve (1.6% of total)
R4	Personnel turnover at field-officer level	Medium	Medium	Y3 cohort hiring under way (open req. F25)
R5	WAADA reporting cycle slip	Low	Medium	Internal review 2 weeks ahead of submission
R6	Field-trial signage / land-use disputes (Udi)	Low	High	MOU with Udi LGA Agric. Extension Office

10. Sustainability & Knowledge Transfer

Programme sustainability rests on three pillars. First, the WAITA Cooperative Federation has committed in-kind co-financing equivalent to ₦12,300,000 across Year-2 in the form of cooperative-level field demonstration days and member outreach. Second, the NNU Office of Sponsored Programmes carries the Year-3 transition budget proposal forward as part of the 2027 institutional planning cycle. Third, programme outputs are released as Creative Commons BY-NC datasets to the WAITA Federation knowledge base, allowing downstream cooperative-level adoption without further licensing friction.

Knowledge transfer to the cooperative tier proceeds through the participatory dissemination programme (Q4 2027 → Q2 2028) with a target of 12 cooperatives across three states. Adoption metrics are tracked in the consortium's Airtable base 'Cassava-Biofort/Cooperative-Adoption-Tracker'.

11. References

- [1] Saltzman, A. et al. (2017) Biofortification techniques to improve food security. Reference Module in Food Science, Elsevier.
- [2] Talwar, D. et al. (2012) AOAC 2005.02 HPLC determination of carotenoids in yellow-fleshed cassava roots. J. AOAC Int. 95(6): 1684–1691.
- [3] IITA (2024) Standard protocol for biofortified cassava field trials. IITA Ibadan Crop Improvement Technical Note 2024-07.
- [4] WAITA Federation (2026) Strategic Plan 2026–2030, §3.2 'Cassava biofortification priority vertical'.
- [5] WAADA (2026) Grant Award Terms & Conditions — WAADA/2026/WAITA-EACRI/Y2/007 (internal ref. WAADA-GT-2026-WAITA-007).

Appendix A — Year-1 Deliverables Status Table

Reproduced verbatim from the consortium Confluence space (WAITA-EACRI / Year-1 Deliverables) as of 2026-09-30 16:42 WAT (snapshot owner: amelia.akpan, WAADA Secretariat). Signage colours match the live workspace.

ID	Deliverable	Owner	Status
D-1-1	Baseline farmer survey (Nsukka LGA, n=340 households)	Layla McBride	DONE
D-1-2	Variety screening Y1 report (4 cultivars × 3 reps)	Amina Bello	DONE
D-1-3	Field-trial plot establishment Udi (n=12)	Layla McBride + Derek Akpan	IN PROGRESS
D-1-4	EACRI laboratory protocol harmonisation	Amina Bello	IN PROGRESS

Appendix B — Letters of Support (Summary)

B.1 WAITA Cooperative Federation — Mr. Tunde Adekoya, Director, Member Services. Letter dated 2026-09-18. Confirms in-kind co-financing of ₦12,300,000 across Year-2 in cooperative demonstration days and member outreach.

B.2 Udi LGA Agricultural Extension Office — Mr. Chidi Anigbogu, Senior Extension Officer. Letter dated 2026-09-22. Confirms continued land-use access for plots UDI-2026-001 through UDI-2026-012 plus the high-fertility control plot UDI-2026-070 through 2028-12-31.

B.3 EACRI Nairobi Office of Research — Dr. Joseph Mbiti, Deputy Director. Letter dated 2026-09-25. Confirms HPLC analytical capacity allocation and cross-site sample-shipment logistics for the Y2 programme period.

Original signed letters are held on file at the NNU Office of Sponsored Programmes and available for WAADA review on request.